Skill - 1

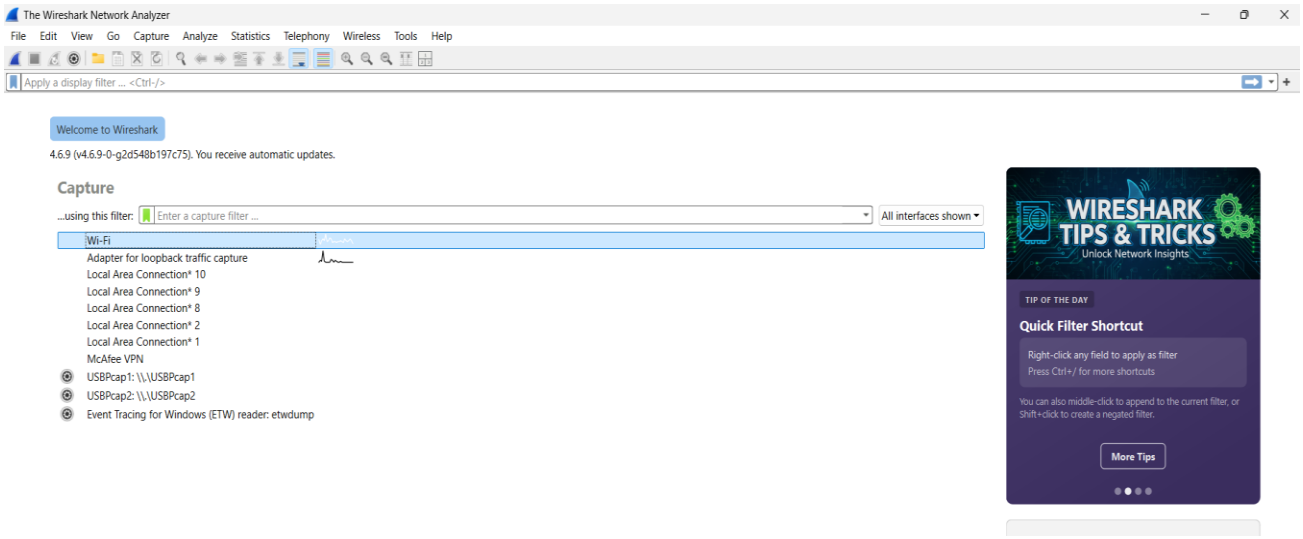
Encapsulation Builder: Analyze the overhead introduced by protocol headers and determine how encapsulation affects network efficiency.

Aim: To capture network packets and understand how Ethernet, IP, and TCP/UDP headers are added during data transmission.

Whenever data travels through a computer network, each protocol layer adds its own header to the data. These headers introduce overhead. Using Wireshark, we will capture real network packets, examine Ethernet, IP and TCP/UDP headers, calculate the total overhead, and determine how efficiently the network carries application data.

Step 1: Install Wireshark

1. Open the browser.
2. Install Wireshark [Wireshark • Go Deep | Download]
3. Install it using the default settings.
4. Open Wireshark.

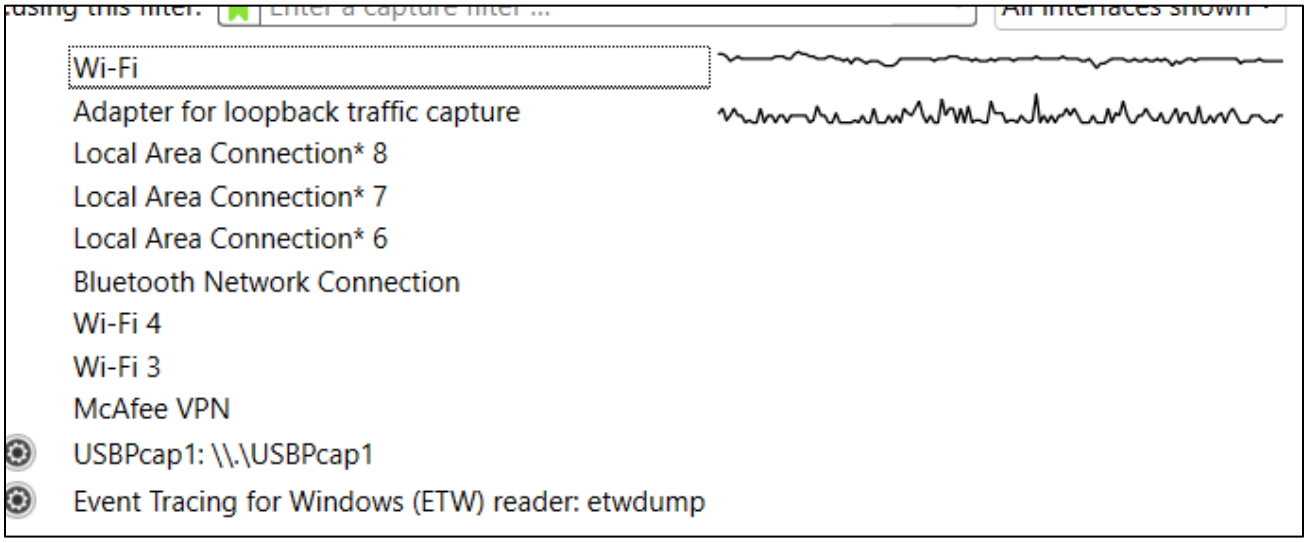


Step 2: Select your network interface

When Wireshark opens, you will see interfaces such as:

Wi-Fi, Ethernet, Bluetooth

1. If you are connected through Wi-Fi, select Wi-Fi.
2. Double-click the Wi-Fi interface.
3. Packet capture will start.



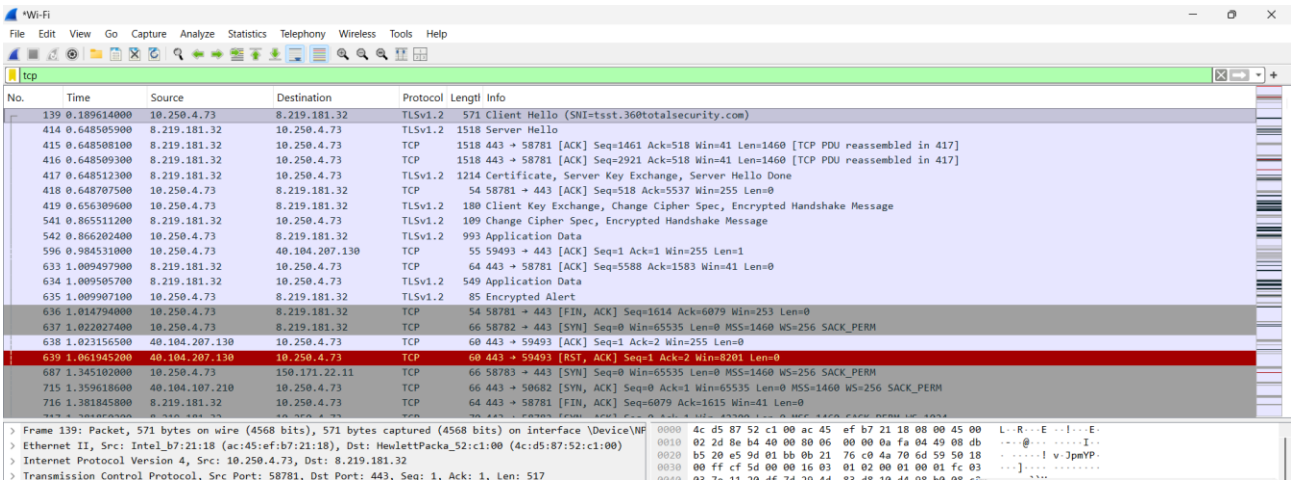
Step 3: Generate network traffic

While Wireshark is capturing:

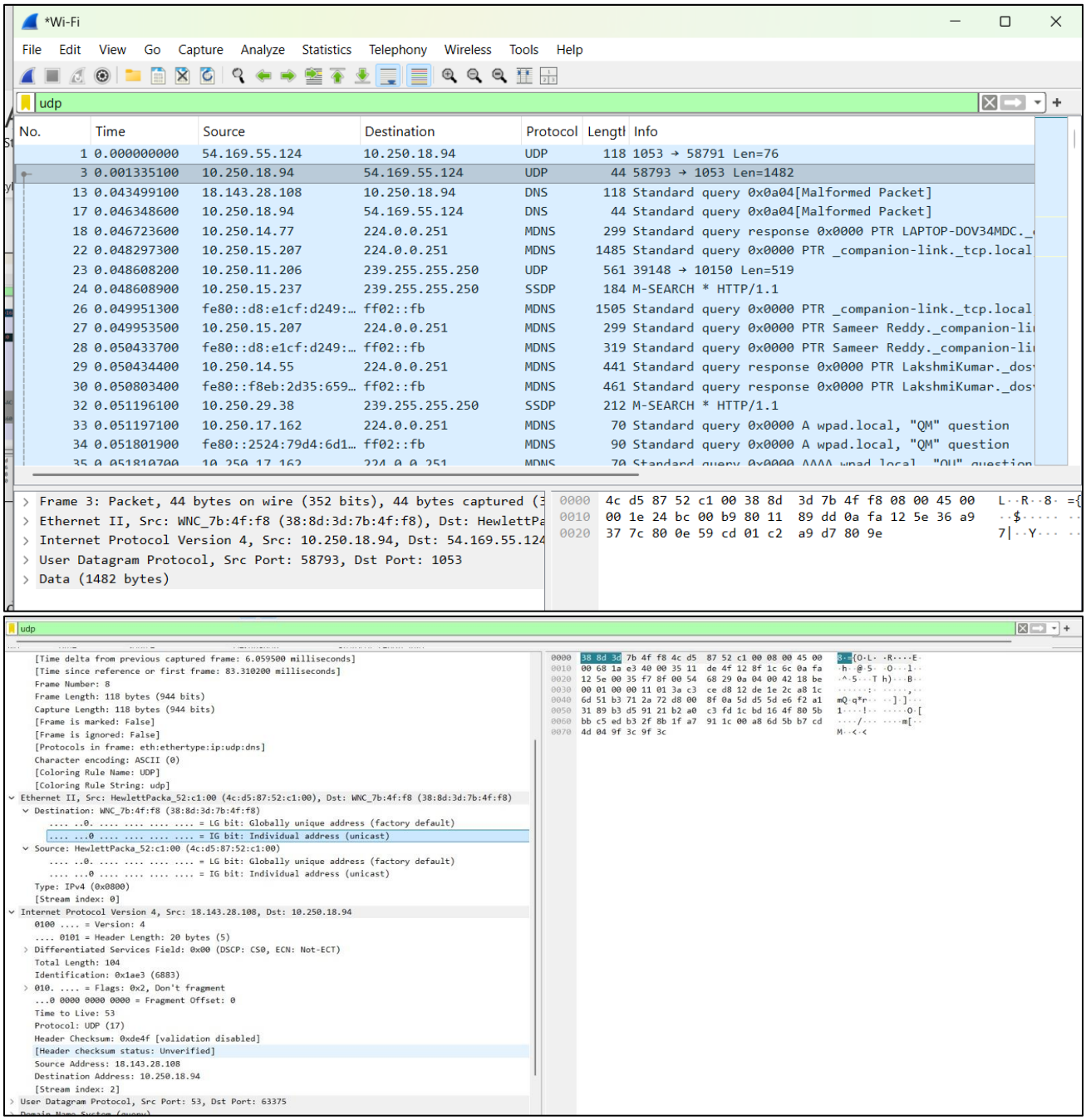
1. Open a new browser tab.
2. Visit a website such as https://www.google.com.
3. Open two or three different websites.
4. Wait for approximately 30 seconds.
5. Return to Wireshark.
6. Click the red Stop button.

You should now see captured packets.

1.TCP

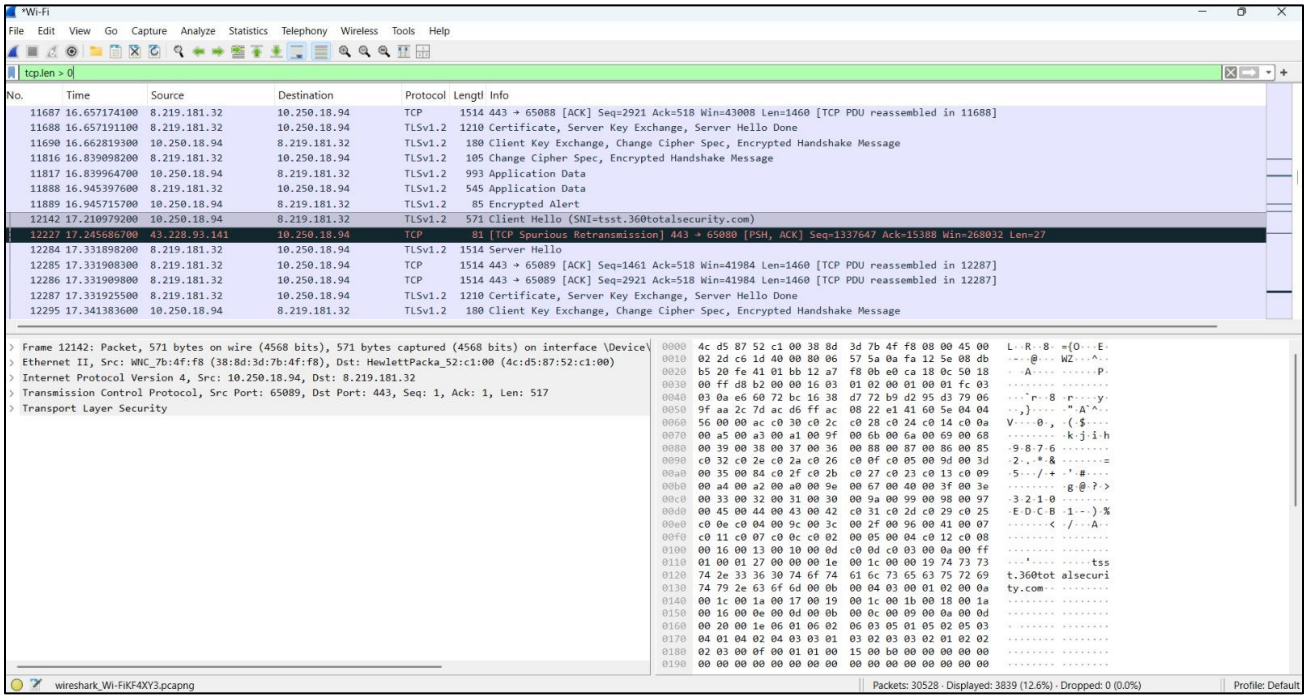


2.UDP

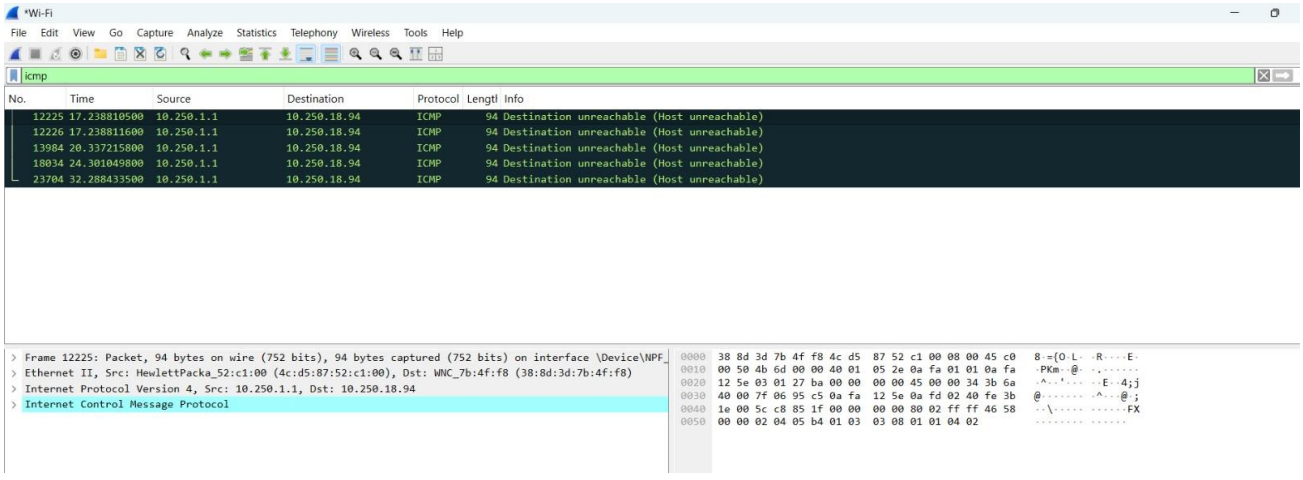

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
No. Time Source Destination Protocol Lengtl Info
1 0.000000000 54.169.55.124 10.250.18.94 UDP 118 1053 → 58791 Len=76
3 0.001335100 10.250.18.94 54.169.55.124 UDP 44 58793 → 1053 Len=1482
13 0.043499100 18.143.28.108 10.250.18.94 DNS 118 Standard query 0x0a04[Malformed Packet]
17 0.046348600 10.250.18.94 54.169.55.124 DNS 44 Standard query 0x0a04[Malformed Packet]
18 0.046723600 10.250.14.77 224.0.0.251 MDNS 299 Standard query response 0x0000 PTR LAPTOP-DOV34MDC._
22 0.048297300 10.250.15.207 224.0.0.251 MDNS 1485 Standard query 0x0000 PTR _companion-link._tcp.local
23 0.048608200 10.250.11.206 239.255.255.250 UDP 561 39148 → 10150 Len=519
24 0.048608900 10.250.15.237 239.255.255.250 SSDP 184 M-SEARCH * HTTP/1.1
26 0.049951300 fe80::d8:e1cf:d249:… ff02::fb MDNS 1505 Standard query 0x0000 PTR _companion-link._tcp.local
27 0.049953500 10.250.15.207 224.0.0.251 MDNS 299 Standard query 0x0000 PTR Sameer Reddy._companion-li
28 0.050433700 fe80::d8:e1cf:d249:… ff02::fb MDNS 319 Standard query 0x0000 PTR Sameer Reddy._companion-li
29 0.050434400 10.250.14.55 224.0.0.251 MDNS 441 Standard query response 0x0000 PTR LakshmiKumar._dos
30 0.050803400 fe80::f8eb:2d35:659… ff02::fb MDNS 461 Standard query response 0x0000 PTR LakshmiKumar._dos
32 0.051196100 10.250.29.38 239.255.255.250 SSDP 212 M-SEARCH * HTTP/1.1
33 0.051197100 10.250.17.162 224.0.0.251 MDNS 70 Standard query 0x0000 A wpad.local, "QM" question
34 0.051801900 fe80::2524:79d4:6d1… ff02::fb MDNS 90 Standard query 0x0000 A wpad.local, "QM" question
Frame 3: Packet, 44 bytes on wire (352 bits), 44 bytes captured (3
Ethernet II, Src: WNC_7b:4f:f8 (38:8d:3d:7b:4f:f8), Dst: HewlettPa
Internet Protocol Version 4, Src: 10.250.18.94, Dst: 54.169.55.124
User Datagram Protocol, Src Port: 58793, Dst Port: 1053
Data (1482 bytes)
0000 4c d5 87 52 c1 00 38 8d 3d 7b 4f f8 08 00 45 00
0010 00 1e 24 bc 00 b9 80 11 89 dd 0a fa 12 5e 36 a9
0020 37 7c 80 0e 59 cd 01 c2 a9 d7 80 9e

udp
[Time delta from previous captured frame: 6.059500 milliseconds]
[Time since reference or first frame: 83.310200 milliseconds]
Frame Number: 8
Frame Length: 118 bytes (944 bits)
Capture Length: 118 bytes (944 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:udp:dns]
Character encoding: ASCII (0)
[Coloring Rule Name: UDP]
[Coloring Rule String: udp]
Ethernet II, Src: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00), Dst: WNC_7b:4f:f8 (38:8d:3d:7b:4f:f8)
Destination: WNC_7b:4f:f8 (38:8d:3d:7b:4f:f8)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Source: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 0]
Internet Protocol Version 4, Src: 18.143.28.108, Dst: 10.250.18.94
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 104
Identification: 0x1ae3 (6883)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 53
Protocol: UDP (17)
Header Checksum: 0xde4f [validation disabled]
[Header checksum status: Unverified]
Source Address: 18.143.28.108
Destination Address: 10.250.18.94
[Stream index: 2]
User Datagram Protocol, Src Port: 53, Dst Port: 63375
0000 38 8d 3d 7b 4f f8 4c d5 87 52 c1 00 08 00 45 00
0010 00 68 1a e3 40 00 35 11 de 4f 12 8f 1c 6c 0a fa
0020 12 5e 00 35 f7 8f 00 54 68 29 0a 04 00 42 18 be
0030 00 01 00 00 11 01 3a c3 ce d8 12 de 1e 2c a8 1c
0040 6d 51 b3 71 2a 72 d8 00 8f 0a 5d d5 5d e6 f2 a1
0050 31 89 b3 d5 91 21 b2 a0 c3 fd 1c bd 16 4f 80 5b
0060 bb c5 ed b3 2f 8b 1f a7 91 1c 00 a8 6d 5b b7 cd
0070 4d 04 9f 3c 9f 3c

3.TCP.LEN > 0



4. ICMP



Step 4: Apply display filters

In the Wireshark filter bar, enter each filter separately and press Enter.

Filter	Purpose

tcp	Displays TCP packets
udp	Displays UDP packets
icmp	Displays ICMP packets
tcp.len > 0	Displays TCP packets carrying data

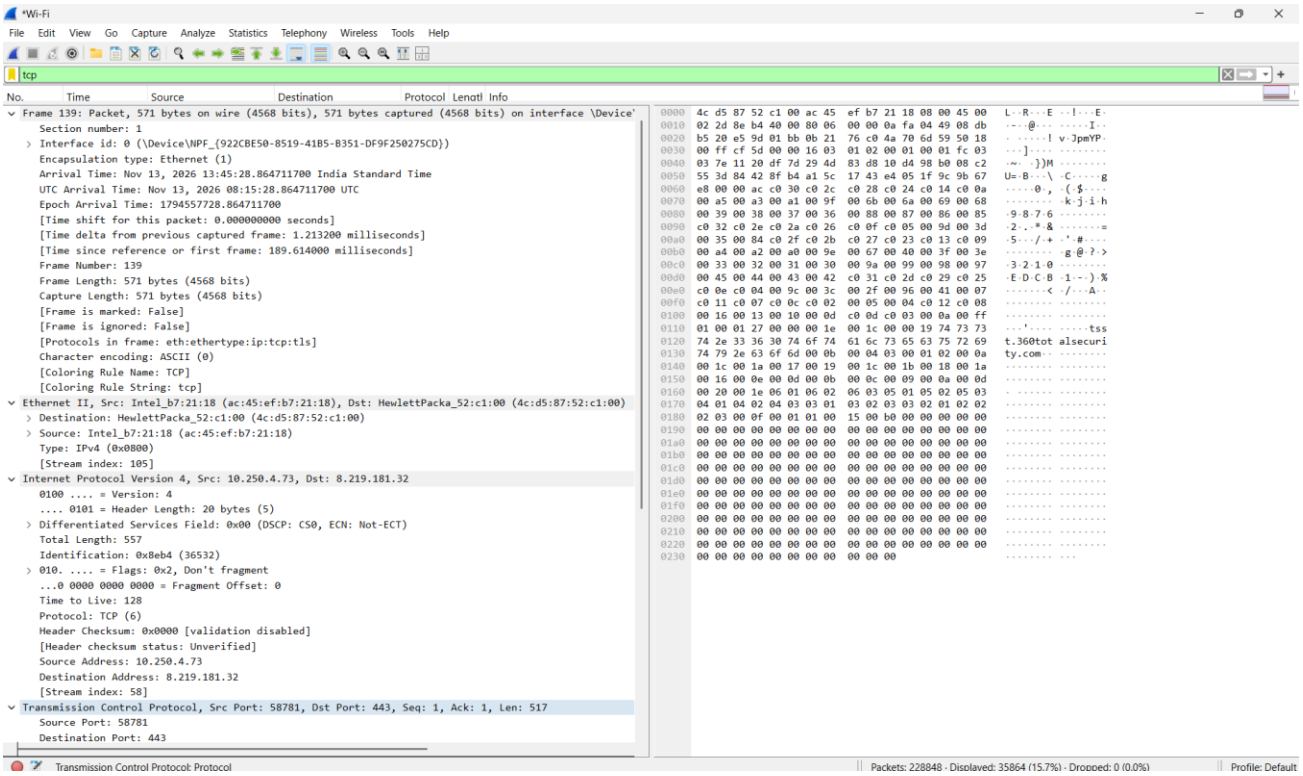
First, enter: TCP
Observe the packets displayed.
Then enter: UDP
Observe the UDP packets.

Step 5: Inspect packet headers

Select any TCP packet from the packet list.

In the packet details pane, expand these sections:

1. Frame
2. Ethernet II
3. Internet Protocol Version 4
4. Transmission Control Protocol



Record the following values:

Parameter	Where to find it
Frame length	Frame
Ethernet Header	Ethernet II
Source MAC Address	Ethernet II
Destination MAC Address	Ethernet II
IPv4 Header Length	Internet Protocol Version 4
IPv4 Total Length	Internet Protocol Version 4
TCP Header Length	Transmission Control Protocol
TCP Segment Length	Transmission Control Protocol

Repeat the process for one UDP packet. Observe the UDP header and its length field.

Step 6: Calculate header overhead

For a basic Ethernet + IPv4 + TCP packet without TCP options:

TCP Packet Calculation – Example 1

Given:

- Ethernet Header = 14 bytes
- IPv4 Header = 20 bytes
- TCP Header = 20 bytes
- TCP Payload = 1000 bytes

Total Packet Size:

14 + 20 + 20 + 1000 = **1054 bytes**

Total Overhead:

14 + 20 + 20 = **54 bytes**

Payload Efficiency:

1000/1054*100 = 94.88%

2. TCP Packet Calculation – Example 2

Given:

- Ethernet Header = 14 bytes
- IPv4 Header = 20 bytes
- TCP Header = 20 bytes
- TCP Payload = 250 bytes

Total Packet Size:

14 + 20 + 20 + 250 = **304 bytes**

Total Overhead:

54 bytes

Payload Efficiency:

250/304*100 = 82.24%

3. UDP Packet Calculation – Example 1

Given:

- Ethernet Header = 14 bytes
- IPv4 Header = 20 bytes
- UDP Header = 8 bytes
- UDP Payload = 500 bytes

Total Packet Size:

14 + 20 + 8 + 500 = **542 bytes**

Total Overhead:

14 + 20 + 8 = **42 bytes**

Payload Efficiency:

500/542*100 = 92.25%

4. UDP Packet Calculation – Example 2

Given:

- Ethernet Header = 14 bytes
- IPv4 Header = 20 bytes
- UDP Header = 8 bytes
- UDP Payload = 50 bytes

Total Packet Size:

14 + 20 + 8 + 50 = **92 bytes**

Total Overhead:

42 bytes

Payload Efficiency:

50/92*100 = 54.35%

Note: Wireshark may not capture or display the Ethernet FCS. Use the actual

captured packet fields when recording your observations.

Step 7:
Fill in your observation table

Record values from your own Wireshark capture.

Parameter	**TCP Packet**	**UDP Packet**
Frame Length	**1054 bytes**	**542 bytes**
Ethernet Header Length	**14 bytes**	**14 bytes**
IP Header Length	**20 bytes**	**20 bytes**
Transport Header Length	**20 bytes**	**8 bytes**
IP Total Length	**1040 bytes**	**528 bytes**
Total Payload Length	**1000 bytes**	**500 bytes**
Total Overhead	**54 bytes**	**42 bytes**
Payload Efficiency	**94.88%**	**92.25%**

Steps to save a capture file

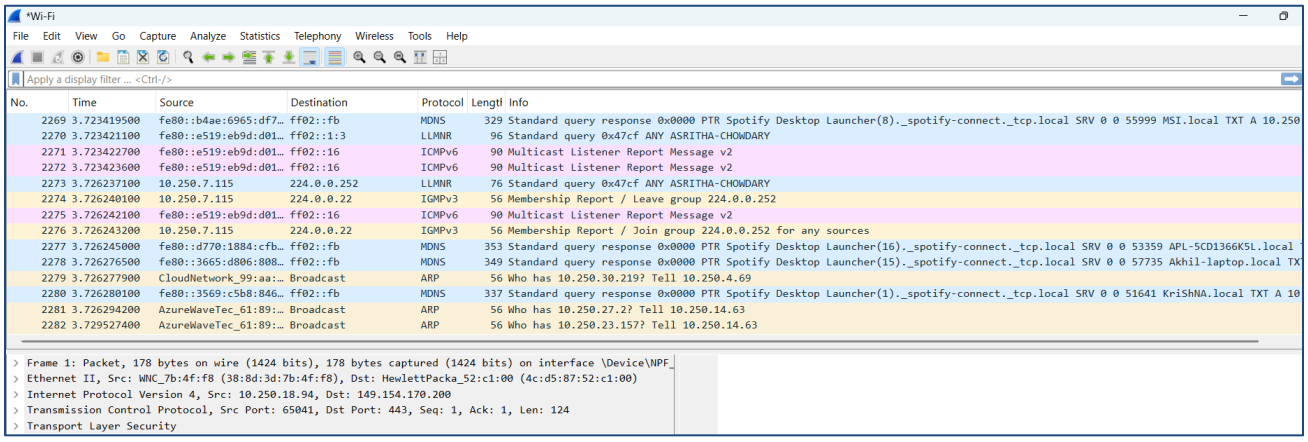
Step 1: Start capturing packets

1. Open Wireshark.
2. Select your Wi-Fi or Ethernet interface.
3. Double-click the interface to start capturing packets.

Step 2: Stop the capture

Once you have generated sufficient network traffic:

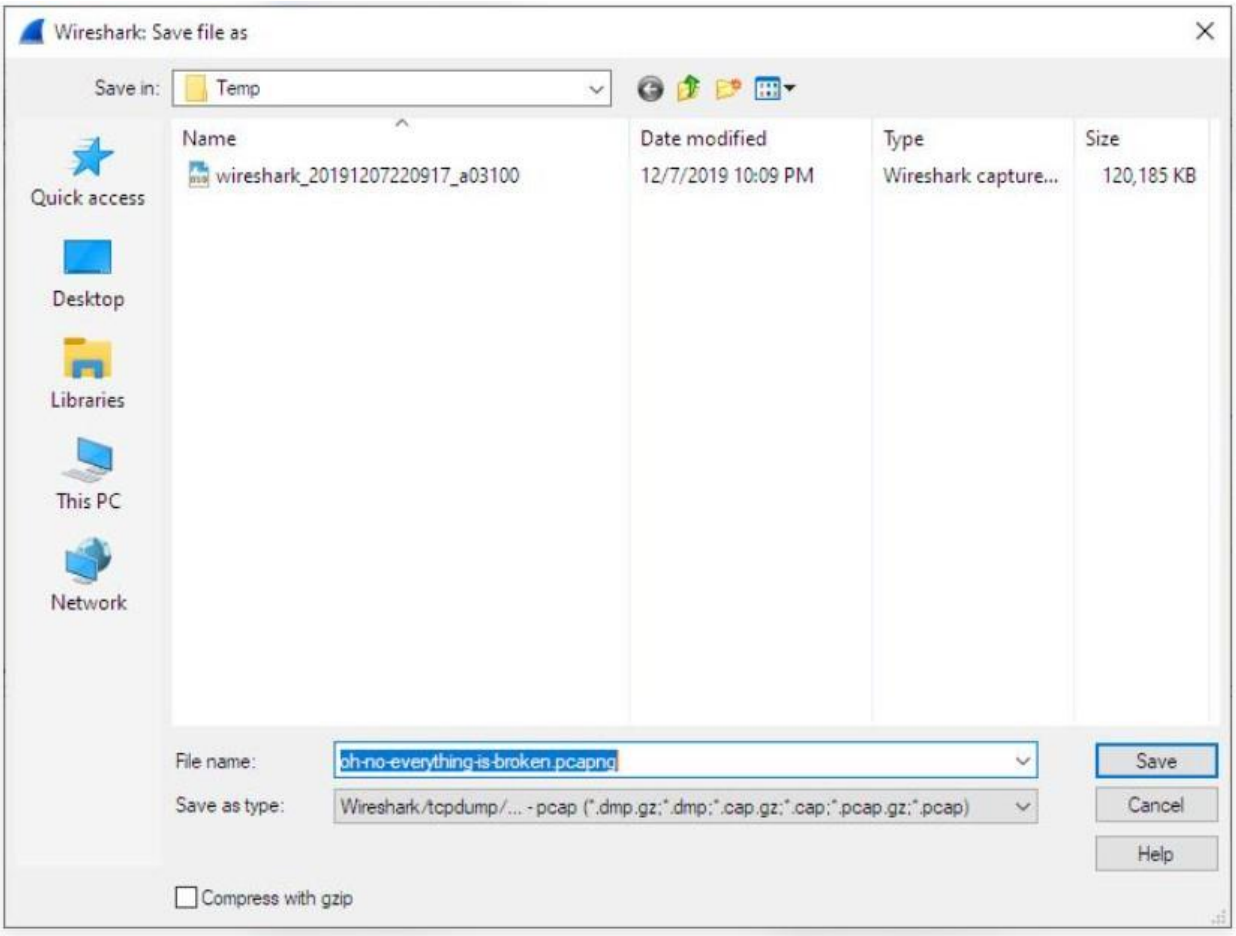
1. Click the red Stop button (top-left corner).
2. Your captured packets will remain visible on the screen.



Step 3: Save the capture

1. Click File in the menu.
2. Select Save As.
3. Choose the folder where you want to save the file.
4. Enter a filename.
5. Keep the file format as pcapng.
6. Click Save.

For example: sender_capture.pcapng



How to open a saved capture file

1. Open Wireshark.

2. Click File → Open.

3. Select your .pcapng or .pcap file.

4. Click Open.

5. Enter icmp in the display filter bar to view ICMP packets.